

1.Explain the Difference between Abstract Class and Interface with example programs

```
abstract class Animal {

    abstract void sound();

    void eat() {
        System.out.println("Animal is eating");
    }
}

interface Pet {
    void play();
}

class Dog extends Animal implements Pet {

    void sound() {
        System.out.println("Dog barks");
    }

    public void play() {
        System.out.println("Dog is playing");
    }
}

public class Main {
    public static void main(String[] args) {

        Dog d = new Dog();

        d.sound();
        d.eat();
        d.play();
    }
}
```

2. Explain about Overloading and Overriding with example programs

```
class Animal {

    // Method Overriding
    void sound() {
        System.out.println("Animal makes sound");
    }
}

class Dog extends Animal {

    // Method Overriding
    @Override
    void sound() {
        System.out.println("Dog barks");
    }

    // Method Overloading
    void eat() {
```

```
        System.out.println("Dog eats food");
    }

    void eat(String food) {
        System.out.println("Dog eats " + food);
    }
}

public class Main {
    public static void main(String[] args) {

        Dog d = new Dog();

        // Overriding
        d.sound();

        // Overloading
        d.eat();
        d.eat("Chicken");
    }
}
```